Deviation register

Deviation register I

A registered report must state, explicitly, every way the executed analysis departed from the frozen plan. This template treats that register as a first-class artifact: `build_deviation_ledger` produces one row per executed outcome and model, classifying each as `ok` (registered), `warning` (an unregistered element carried with a documented rationale), or `error` (an unregistered element with no rationale). `compare_analysis_to_registration` downgrades a documented model change from `error` to `warning` and refuses to accept any deviation whose `rationale` is empty.

Demonstration deviations I

To exercise the machinery, the demonstration executes a deliberately plan-divergent analysis: it adds a `secondary_score` endpoint and swaps the registered `permutation_test` for a `linear_model`, each accompanied by a rationale. The resulting ledger is:

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Order	Kind	Target	Registered	Documented	Severity
0	outcome	primary_score	yes	—	ok
1	outcome	secondary_score	no	yes	warning
2	model	linear_model	no	yes	warning

Demonstration deviations I

Because both departures are documented, the review packet remains valid: it reports `primary_score` as the only confirmatory outcome, `secondary_score` as exploratory, and an integrity score of `0.9` (a single `0.10` warning penalty against a perfect plan). Had either deviation lacked a rationale, the same code would have raised a blocking `deviation_without_rationale` or `unregistered_outcome` error, and the packet would be invalid. fig. 4 plots this register along the registered-to-executed axis.

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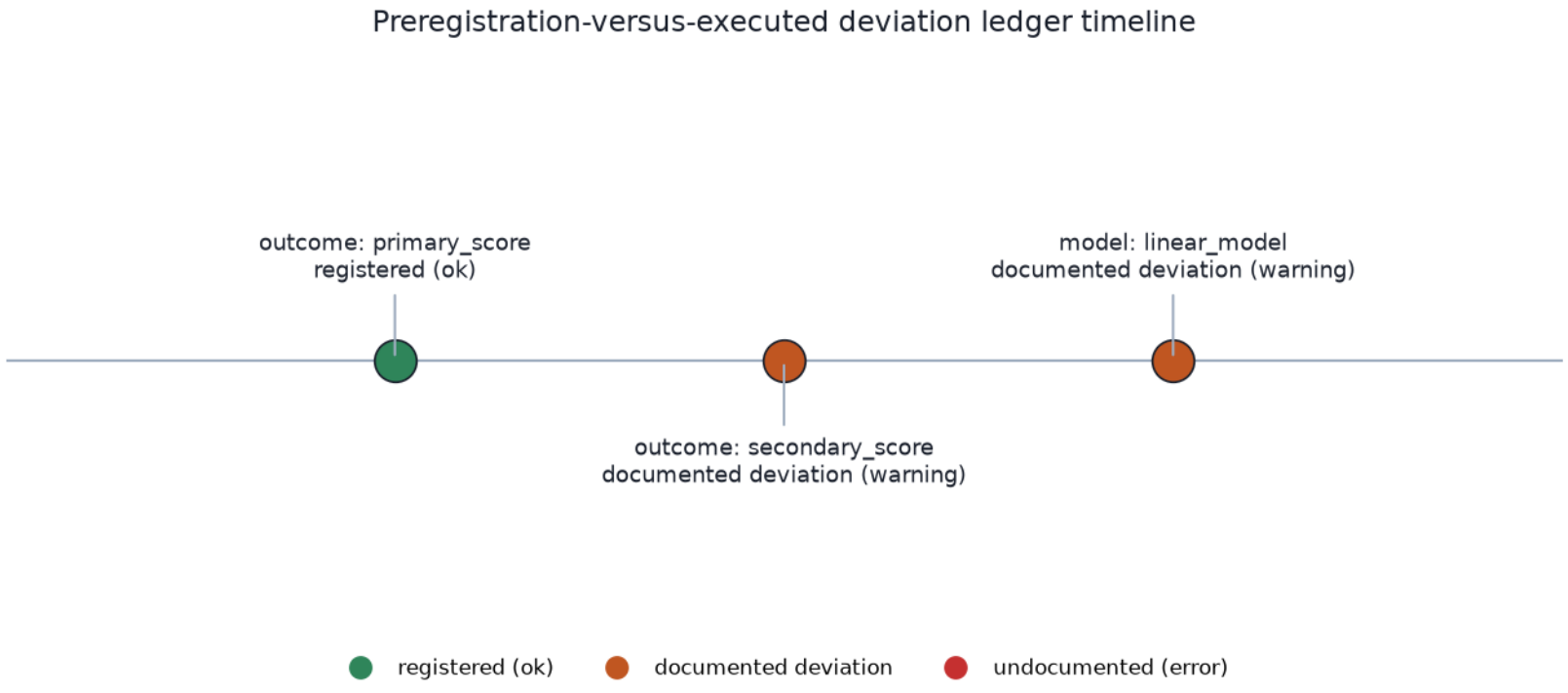

Figure 4: Preregistration-versus-executed deviation ledger timeline, rendered by `plot_deviation_timeline` from `build_deviation_ledger` and `deviation_timeline`. Each marker is a ledger row coloured by severity: the registered `primary_score` outcome is green (`ok`), while the added `secondary_score` endpoint and the `linear_model` substitution are amber (`warning`, documented deviations). No marker is red because every departure

Demonstration deviations I

carries a rationale. An undocumented departure would appear in red and invalidate the review packet.

Reading the register I

The register is the mechanism that keeps Section 5 honest: the confirmatory claim is restricted to the registered `primary_score` outcome, and the exploratory `secondary_score` endpoint is visible, labelled, and excluded from confirmatory interpretation. Forks should extend this ledger — never edit the frozen registration — whenever the executed analysis must diverge from plan.